

DEMO WALKTHROUGH

Clinical Dashboard Demo — Explanation Guide

What the interactive prototype shows, how to read each panel, and what is simulated versus real.

1. What this demo is

This is a single-page, front-end-only prototype of the doctor-facing dashboard — the "Output / Visualization" step described in Phase 3 of the system architecture. It runs entirely in the browser with synthetic, hand-written data: there is no backend, no database, and no real patient information behind it.

Its job is to validate the layout and interaction model — where the SOFA score lives, how the time window affects everything on screen, how a radiology finding sits next to lab trends — before that layout is wired to the real Phase 1–3 pipeline (Blob Storage, PostgreSQL, the vector DB, and Azure OpenAI).

2. Screen layout

2.1 Top bar

- A patient switcher on the right lets you jump between the three demo patients described in Section 3. Each button shows the patient's first name and their scenario badge.

2.2 Left column

- **Upload Reports:** a drag-and-drop area for PDF or image files. Dropped or browsed files are listed underneath with a remove control. This stands in for Phase 1, Step 1 (Upload via Web App/API) — files are kept in memory for the session only and are not extracted or stored.
- **Time Window:** a dropdown for Last 6 / 12 / 48 hours. It filters which vitals readings and lab readings are shown everywhere else on the page — the vitals table, every trend chart, and the progress summary all react to this one control.
- **Progress Summary:** a short list comparing the earliest reading in the selected window against the latest, per test, with an up/down/flat icon. For the septic patient it also states how the SOFA total has changed. This is where Phase 3's LLM-generated summary would eventually live — in the demo it's a deterministic template over the visible numbers, not a live model call.

2.3 Center column

- **Patient header:** name, scenario badge, diagnosis, MRN, age/gender, bed, attending physician, admission time, and length of stay.
- **SOFA Score card (conditional):** only appears when the selected patient is flagged septic. Shows the current total against the previous total, a severity band (e.g. "Moderate organ dysfunction"), and all six component sub-scores — Respiration, Coagulation, Liver, Cardiovascular, CNS, Renal — each on the standard 0–4 scale, with the prior value shown alongside for comparison.
- **Vitals table:** deliberately horizontal: each column is a timestamp within the selected window, each row is a vital sign (Heart Rate, Blood Pressure, Respiratory Rate, SpO₂, Temperature) — matching the layout you asked for.
- **Report Comparison:** a grid of small line charts, one per lab test (WBC, Hemoglobin, Platelets, Creatinine, CRP, Lactate). Each chart shades its normal reference range and flags whether the latest value is trending into or out of that band.

- **Radiology section (conditional):** only appears for the patient with an imaging report. Shows an illustration alongside the Findings and Impression text pulled from that report.

3. The three demo patients

Each patient exercises a different combination of the conditional sections, per your original spec:

Patient	Scenario & what it demonstrates
Robert Chen	Septic — the only patient where the SOFA Score card renders. Data is written to show 48-hour improvement (SOFA 13 → 4) alongside falling lactate, WBC, and creatinine, so the Progress Summary reads as a clear recovery story.
Maria Gomez	Standard (non-septic) — the SOFA card is hidden entirely. Demonstrates the layout with that section absent. Vitals and labs are low-acuity and steady, typical of routine post-operative monitoring.
David Okafor	Radiology — SOFA hidden, radiology section shown. Demonstrates the imaging-plus-text layout: a chest X-ray illustration next to Findings/Impression describing improving pneumonia.

4. Data & assumptions

Read this before treating anything on screen as real

- All three patients, names, and values are fully synthetic — written for this demo, not drawn from any real record.
- Timestamps are generated relative to a fixed reference time so the 6/12/48-hour filters have something meaningful to show.
- The radiology image is an original schematic illustration, explicitly labeled as such on screen — not an actual X-ray or any real medical image.
- The Progress Summary is a deterministic first-vs-latest comparison over the values on screen, not an LLM call.
- SOFA component scores are hand-set to tell a coherent, improving story. They are not computed by a validated calculator and should not be read as clinical output.

5. How this maps to the system architecture

For reference against the architecture document, here is which piece of the demo stands in for which phase/step:

Demo element	Architecture phase / step it represents
Upload Reports panel	Phase 1, Step 1 — Upload via Web App / API
Time Window + Vitals table	Phase 3, Steps 1–2 — retrieve vitals from SQL, analyze trends
Report Comparison charts	Phase 3, Steps 1–2 — retrieve lab reports from SQL, compare current vs. previous
Radiology section	Phase 3, Step 1 — retrieve radiology images from Blob Storage and text from the Vector DB
Progress Summary panel	Phase 3, Steps 3–4 — LLM summary / Generate Progress Notes (currently simulated client-side)

Demo element	Architecture phase / step it represents
The page as a whole	Phase 3, Step 5 — Output / Visualization, the web dashboard for doctors

6. Not yet wired up

- No real upload processing — no Azure Function trigger, no Document Intelligence extraction.
- No live PostgreSQL, Vector DB, or Blob Storage — every value is embedded directly in the demo's source.
- No real Azure OpenAI call generating the Progress Summary.
- No authentication or role-based access control.

These are exactly the gaps the architecture document's Phases 1–3 fill in — this demo is the target shape of the screen those phases feed.